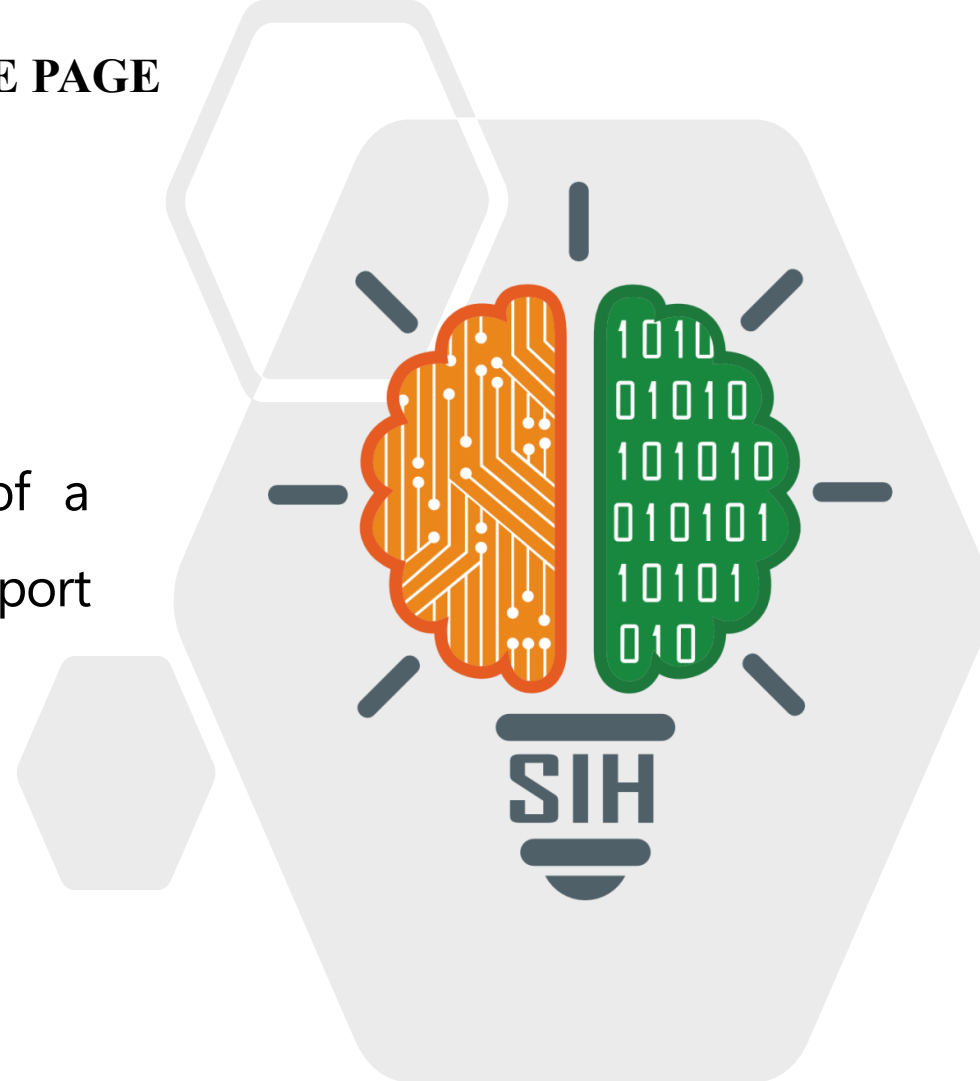


SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID** – 25092
- **Problem Statement Title-** Development of a Digital Mental Health and Psychological Support System for Students in Higher Education
- **Theme-** MedTech / BioTech / HealthTech
- **PS Category-** Software
- **Team ID-**
- **Team Name -** SAHARA



IDEA TITLE



❖ Proposed Solution

A secure platform designed to address student mental health by integrating confidential support tools with university oversight. It empowers students with self-assessments, 24/7 AI support, and seamless doctor access, while providing administrators with anonymized, aggregated data to improve campus-wide wellness initiative.

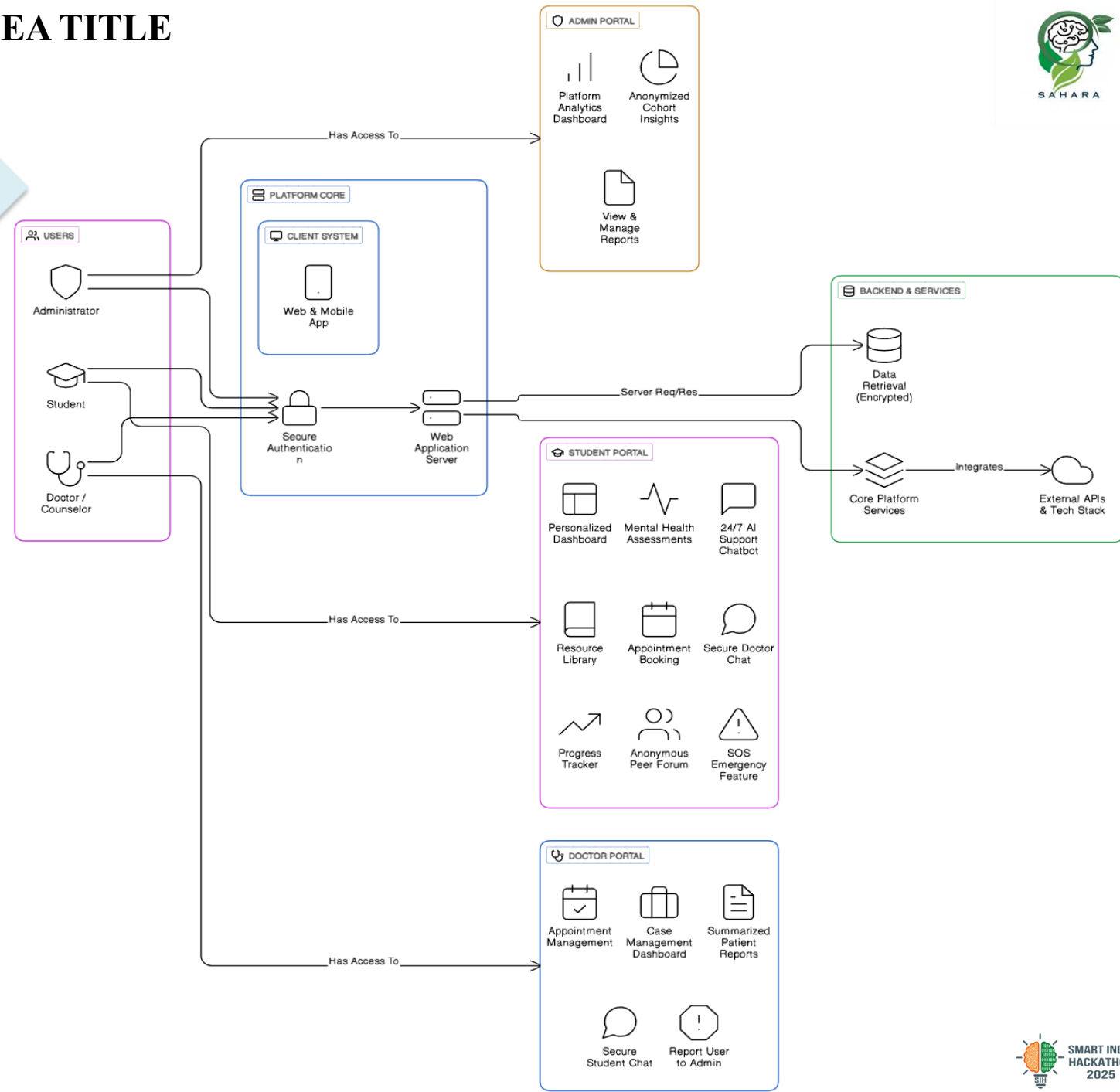
Secure & Anonymous Onboarding: Students register with their official college ID, but the system generates a **unique, anonymized login ID** for all platform use, ensuring complete privacy and **removing the fear of stigma**

Guided Mental Health Assessments: The platform provides self-assessments for anxiety and depression that categorize results **into Low, Moderate, or Severe**. Based on the severity, it recommends a clear pathway keeping a **weekly/monthly data of that student..**

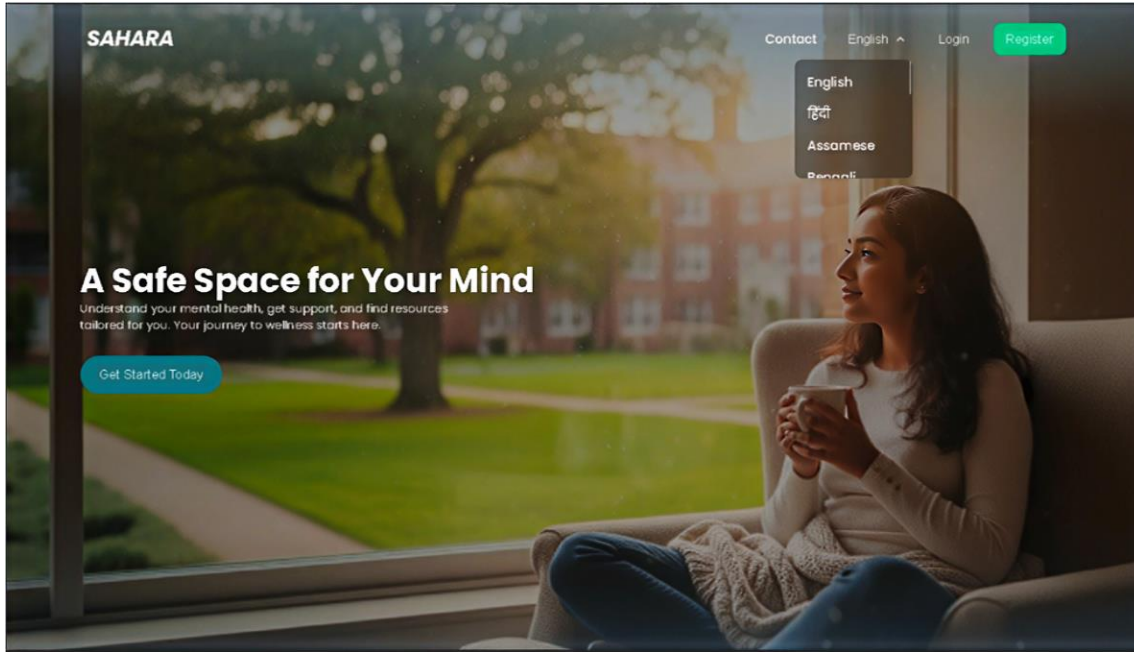
24/7 AI-Powered Support: **An empathetic AI chatbot** is available around the clock for immediate support. **It retains conversation history** to provide personalized, contextual, and continuous assistance **throughout the student's journey**.

Integrated Professional Care: Students can seamlessly **book online or offline appointments** with university-affiliated doctors via a simple calendar system. For emergencies, a dedicated **SOS button** provides an **instant connection to government-assigned doctors** for crisis intervention.

Actionable, Anonymized University Insights: The **admin portal** displays aggregated, anonymized data on the stress and anxiety levels of specific student cohorts, like **a particular department or academic year**. This allows the university **to identify high-pressure areas** and deploy targeted support without compromising individual student privacy



TECHNICAL APPROACH



Sahara: Technical Architecture Overview

Our platform is a modern microservice application with four key layers:

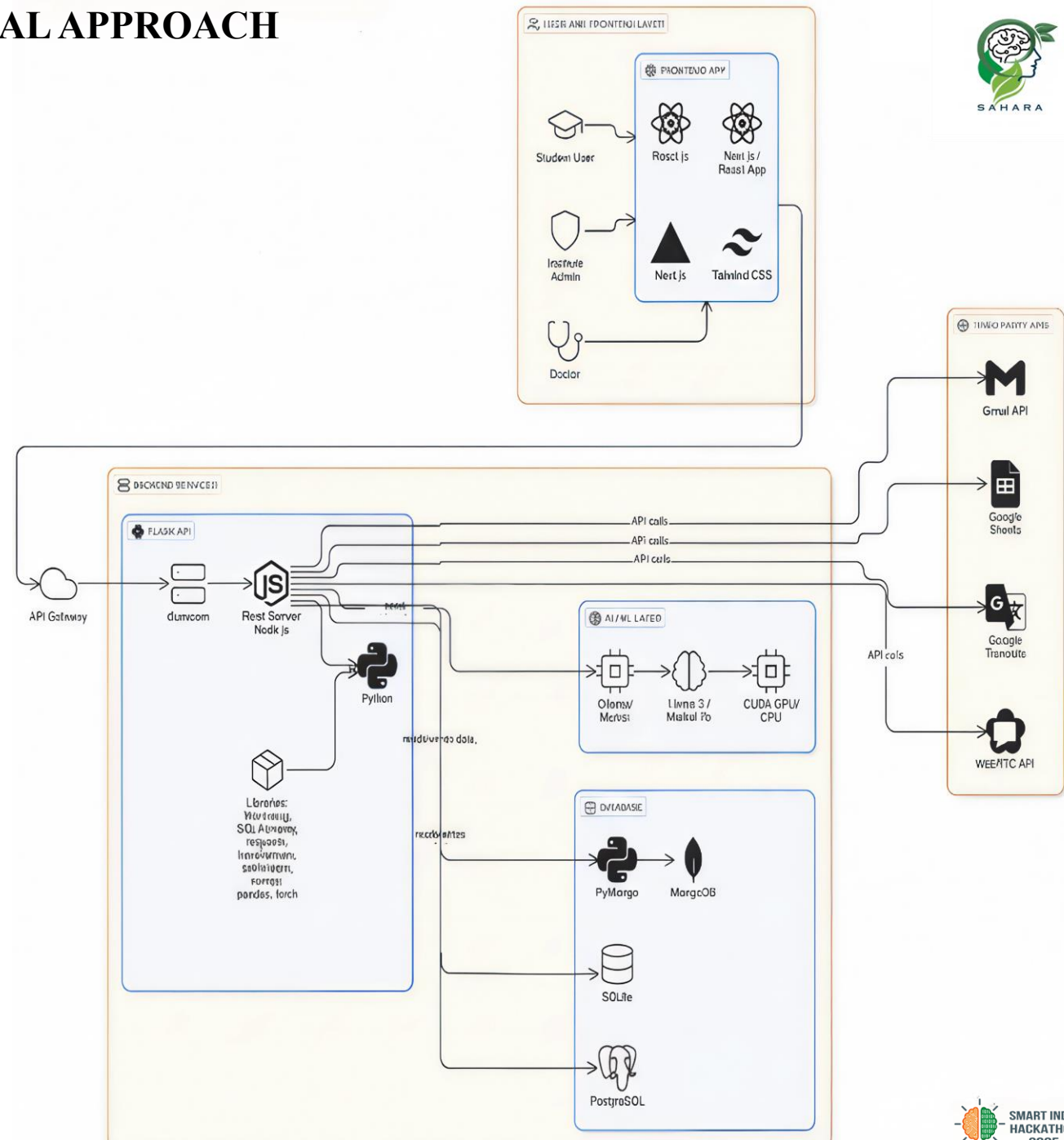
Frontend: A responsive user interface for Students, Admins, and Doctors, built with **Next.js, React and Tailwind CSS**

Backend: The core logic is handled by multiple services:

- **Node.js:** Manages primary API requests and business logic.
- **Python:** Powers data science and machine learning tasks.
- **AI/ML Service:** A dedicated, GPU-accelerated service runs large language models like **Llama 3**.

Database: A hybrid storage approach using **MongoDB** for flexible data and **PostgreSQL** for structured data.

Integrations: We leverage external services like **Google APIs** (Gmail, Sheets, Weather) for enhanced functionality and **WebRTC** for real-time communication.



❖ Key Challenges:

- Data Privacy and Security – Ensuring that sensitive data is securely stored and **protected from breaches**.
- User Engagement – Students may be **hesitant** to use digital platforms **due to stigma around mental health**.
- Accuracy of Support – Providing **accurate, reliable, and ethical psychological support** is critical.
- Technical Challenges – Maintaining uptime, high traffic, and **integrating with existing university systems**.
- Complex Algorithm – Most of the modern apps have a subscriptions, that blocks useful features behind **pay walls**, let alone the **complex navigation and regional language barrier**.

❖ Strategies for Overcoming:

- Data Privacy and Security – Ensuring that sensitive data is securely stored and **protected from breaches**.
- User Engagement – Students may be **hesitant** to use digital platforms **due to stigma around mental health**.
- Accuracy of Support – Providing **accurate, reliable, and ethical psychological support** is critical.
- Technical Challenges – Maintaining uptime, high traffic, and **integrating with existing university systems**.
- Complex Algorithm – Most of the modern apps have a subscriptions, that blocks useful features behind **pay walls**, let alone the **complex navigation and regional language barrier**.

Supporting Facts

- According to the World Health Organization (WHO), nearly 15% of adults aged 18–29 suffer from mental health disorders, which directly includes the student population.
- A survey conducted by the Indian Journal of Psychiatry highlights that 1 in 4 students in higher education experience symptoms of anxiety and depression.
- A study by the All-India Survey on Higher Education (AISHE) indicates that over 38 million students are enrolled in Indian higher education, showing the large target population.
- Digital interventions for mental health have shown an improvement of 30–40% in coping mechanisms when compared to no intervention

IMPACT AND BENEFITS



SAHARA

Social Benefits

Reduces Stigma & Increases Access by offering a confidential 24/7 AI chatbot and an anonymized login ID, providing a private first step to seeking help.

Fosters Community & Reduces Isolation through a planned anonymous, moderated peer support forum where students can connect over shared experiences.

Provides Culturally & Linguistically Relevant Support with a multilingual interface, addressing the need for a non-generic solution tailored to diverse student bodies.

Educational Benefits

Enhances Student Resilience & Coping Skills by delivering AI-guided coping strategies in chat and providing a curated psychoeducational Resource Library.

Improves Academic Performance by addressing underlying mental health challenges like academic stress and burnout, which directly impact student focus and success..

Enables Proactive & Early Detection using standardized assessments (PHQ-9/GAD-7) and AI-powered chat analysis to identify at-risk students before a crisis.

Economical Benefits

Provides Cost-Effective Support at Scale as the 24/7 AI chatbot handles a high volume of initial support requests, reducing the load on human counselors.

Improves Student Retention & Success by addressing mental health issues that contribute to academic withdrawal, protecting both student investment and institutional enrollment.

Optimizes Institutional Resource Allocation by using the Admin Dashboard's anonymized analytics to deploy wellness resources to specific, high-need student cohorts.

Innovation

Advanced AI-Guided Intervention is achieved through a state-of-the-art LLM that provides nuanced understanding, backed by a crisis detector for safety.

Provides a Continuous & Personalized Care Journey as the database retains conversation history and tracks assessment scores, allowing the AI to offer truly continuous and context-aware support over time.

Reduces Stigma & Increases Access by offering a confidential 24/7 AI chatbot and an anonymized login ID, providing a private first step to seeking help.

RESEARCH AND REFERENCES



➔ Reference and Research Works:

a. Journal:

1. Digital Mental Health Interventions for University Students: A Systematic Review and Meta-Analysis by Harrer, M., Adam, S. H., Baumeister.

b. Article:

1. Prevalence of Depression among University Students: A Systematic Review and Meta-Analysis Study by Diana Sarokhani.
2. Artificial Intelligence-Based Chatbots in Promoting Mental Health and Well-Being: A Systematic Review by : Abd-Alrazaq, A. A, Alajlani, Bewick.

c. Links:

1. <https://repository.gheli.harvard.edu/repository/1269/>
2. <https://www.scu.edu/ethics/focus-areas/bioethics/resources/ai-therapist-data-privacy.html>
3. <https://www.irejournals.com/formatedpaper/1706902.pdf>
4. <https://ijrpr.com/uploads/V6ISSUE4/IJRPR42582.pdf>

How our solution stands compared to others:

Features	SAHARA	Wysa	Youper	Talk-Space	7-cups
Free	✓	✗	✗	✗	◐
User Intent based AI chatbot	✓	◐	◐	✗	✗
24/7 Medical and AI support	✓ On-campus university counselors and free audio/video sessions	✓	✗	✗	✗
Emergency SOS/Crisis detection	✓	◐	✗	✗	✗
Clinical Assessments	✓	◐	◐	✗	✗
Institutional Integration & Admin Tools	✓	✗	✗	✗	✗
Ease of access/ navigation	✓ Multilingual support and easy to use	◐	✗	✗	✗

✓ Full compliance ◐ Partial compliance ✗ Non-compliance/buggy